Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam E · Form 2: Pulley Focus

Multiple Choice Questions: Select the best answer. (1 point each)

1. A cart–hanger graph of acceleration versus hanger weight has a nonzero negative intercept at fixed system mass. Which observation supports resistance rather than a constant acceleration sensor bias?

(A) The graph has many samples at one setting

(B) Both axes are labeled

(C) Independent zero-acceleration calibration is correct and a force probe detects opposing drag

(D) The line is straight

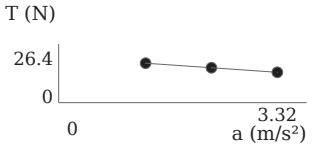
2. An unseen hanger rides on one side of an ideal pulley. Vary the opposite load and record tension with the unseen hanger acceleration positive downward. Use g=10 m/s². Replace the unseen hanger by twice its mass at the same accelerations. What is the new intercept minus the original intercept?

(A) 28.74

(B) 31.89

(C) 30

(D) 30.63



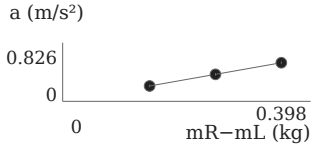
3. Transfer mass between two sides of an ideal Atwood pulley, keeping total mass 5 kg. Right downward is positive; g=10 m/s². mR-mL (kg) -> a (m/s²): (0.1584, 0.3168); (0.2784, 0.5568); (0.3984, 0.7968). With vertical error bounded by 0.0288 and an exact model, what is the error bound on an inferred X?

(A) 0.0144

(B) 0.0744

(C) 0.1344

(D) -0.0456



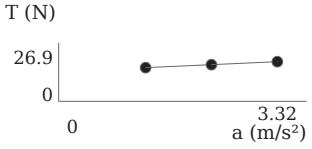
4. The left mass on an ideal pulley is 2 kg; vary the right mass. Left upward is positive; use g=10 m/s². Double the left mass and record tension at the same acceleration values. At the same X=2.32, what is the new minus old vertical reading?

(A) 25.9

(B) 24.64

(C) 25.06

(D) 23.8



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Transfer mass between a cart and hanger at fixed total mass against constant friction. Hanger weights 2 N and 4 N give accelerations 0.2 and 0.6 m/s².

a. Mark the cart and hanger as one system and identify tension as internal. **(1 pt)**



b. Infer mass and resistance; identify which estimate a constant acceleration bias changes. **(3 pt)**

6. An incline block of weight 10 N rests on a 30° ramp with a 5 N hanger on an ideal pulley. Add vertical downward force P to the block. Friction is sufficient; uphill is positive. P (N) -> fs signed (N): (1.32, 0.66); (2.32, 1.16); (3.32, 1.66).

a. Build the model for pressing on an incline from physics and predict both slope and intercept. **(2 pt)**

b. Change the tilt so sinθ increases by a factor of 1.5. Static friction remains sufficient. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A motorless cabin of mass M accelerates upward at a measured value a with a heavier counterweight H. Derive H from the measurement and state why a must be less than g. **(3 pt)**